

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Produced detailed **3D engineering drawings using SolidWorks** for bespoke architectural metalwork, supporting a **15% increase** in design output for the team.
- Assisted sales and estimating teams with **material take-offs** and preliminary design reviews, contributing to the preparation of customer quotations for key projects.
- Generated and managed **works orders and bills of materials** for Statii ERP, ensuring accurate production documentation aligned with quality procedures.
- Maintained detailed **batch traceability and drawing revision records** for fabricated steel and aluminum components, adhering to company quality standards.
- Recorded and allocated engineering and design hours accurately to projects, supporting commercial tracking and enhancing **operational efficiency**.
- Liaised directly with customer design managers and architects to coordinate technical requirements and resolve design queries, facilitating smoother project delivery.
- Contributed to continuous improvement initiatives within the design office, optimising document control and exploring **design automation opportunities**.

SKILLS

CAD Software – SolidWorks, AutoCAD

ERP Systems – Statii, ERP

Engineering Disciplines – Mechanical Engineering, Product Design, Manufacturing Engineering, Architectural Technology, Metalwork Fabrication

Technical Documentation – 2D Drawings, 3D Models, Bills of Materials, Works Orders, Material Take-offs, Batch Traceability

Business Skills – Estimating, Costing, Quotations, Project Management, Commercial Tracking, Operational Efficiency

PROJECTS

Bespoke Metalwork Design System (HCL Technologies)

- Developed a standardised library of components and repeatable design systems within **SolidWorks** for modular architectural metalwork, reducing design time for common elements by **20%**.
- Generated detailed **3D models and 2D fabrication drawings** for custom steel and aluminum projects, ensuring adherence to strict tolerances and client specifications.
- Collaborated with production teams to refine designs based on fabrication feedback, improving manufacturability and reducing material waste by approximately **8%**.

Automated Bill of Materials Generation (HCL Technologies)

- Engineered a Python script to automate the generation of **Bills of Materials (BOMs)** from 3D CAD models for custom fabrication projects, reducing manual data entry by **75%**.
- Integrated the script with existing tracking systems to ensure accurate material allocation and support **batch traceability** requirements.
- Designed a user-friendly interface for the script, enabling non-technical staff to initiate BOM generation, thereby improving workflow efficiency across the design office.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)